

Durgasantosh G

📍 Dallas, TX

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SUMMARY

Software Engineer with 3+ years of experience building responsive and high-performance web applications using **React.js** and **Spring Boot**. Strong in developing reusable UI components, managing **state management** and **asynchronous data flows**, and integrating secure **RESTful APIs** within **microservices-based architectures**. Experienced in improving **frontend performance**, resolving **production issues**, and delivering reliable applications deployed on **AWS** through **CI/CD pipelines**. Built an **AI-powered chatbot using Retrieval-Augmented Generation (RAG)** with a React frontend and FastAPI backend.

EDUCATION

University of North Texas

Masters of Science – Computer Science

Denton, Texas

August 2023 – May 2025

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, C++, C#,SQL, HTML5, CSS3

Frontend: React.js, React Hooks, Angular, Bootstrap 5, jQuery, AJAX, JSON, XML, Redux, GraphQL

Java & Spring: Spring Boot, Spring MVC, Spring REST, Spring Security, Spring Batch, Spring ORM

Microservices & Messaging: Microservices Architecture, RESTful APIs, Apache Kafka, RabbitMQ, Java Multithreading

Cloud & DevOps: AWS (EC2, S3, RDS, SQS, SNS, CloudWatch), Docker, Kubernetes, Jenkins, Git, Maven, Postman

AI & Generative AI: AI Agents, Retrieval-Augmented Generation (RAG), LLM Orchestration, SentenceTransformers

Databases: Oracle, PostgreSQL, MySQL, SQL Server, DB2, MongoDB, Vector Databases (ChromaDB)

Tools & OS: IntelliJ IDEA, Eclipse, VS Code, Linux

EXPERIENCE

Citi Group – New Jersey, USA

May 2025 – Present

Software Engineer

- Developed and enhanced user-facing web applications using **React.js**, building reusable, responsive UI components for enterprise financial platforms.
- Improved frontend performance by approximately **25%** through optimized API consumption, memoization, efficient state management, and reduction of unnecessary re-renders.
- Implemented client-side validation, dynamic form handling, and conditional rendering logic to improve usability and reduce backend validation errors.
- Designed and integrated React components with **Java 17** and **Spring Boot** microservices, ensuring seamless frontend-backend communication and consistent data handling.
- Developed and maintained RESTful APIs, implementing validation, exception handling, and secure access control using **Spring Security** and JWT-based authentication.
- Worked with **Kafka** and **DynamoDB** to support reliable backend data processing and low-latency updates for UI-driven workflows.
- Participated in API contract discussions and ensured backward compatibility to minimize frontend regressions during releases.
- Diagnosed and resolved production issues across both frontend and backend layers, improving application stability and reducing recurring defects.
- Contributed to CI/CD pipelines and containerized deployments using **Docker**, **Kubernetes**, and **Jenkins/GitLab CI**, enabling faster and more stable releases.

Zensar Technologies – India

June 2021 – July 2023

Software Developer

- Worked on end-to-end feature development across **Angular 15**, **React.js**, and **Spring Boot** within a microservices-based financial application.
- Built reusable UI components and implemented client-side form validations and dynamic data handling, helping reduce repetitive code and improving development speed by approximately **30%**.
- Integrated frontend modules with REST APIs, managing asynchronous API calls and handling loading states and error responses to ensure smooth user interaction.
- Developed RESTful APIs using **Java** and **Spring Boot**, implementing input validation, exception handling, and secure integration between services.
- Assisted in implementing Kafka-based workflows to support asynchronous processing for transaction-related backend operations.
- Contributed to PCI DSS-compliant financial modules, helping reduce transaction processing errors by around **20%** through improved validation and error handling logic.
- Participated in AWS deployments and CI/CD automation using **Jenkins** and **Docker**, supporting stable releases within Agile development cycles.

ACADEMIC PROJECTS

Beyond Sight: Empowering Visually Impaired Users with Audible Graphs – UNT Denton

[Link for Research Paper](#)

- Created and deployed an **AI-powered accessibility system** that converts **Excel charts** and **image-based graphs** into audio narratives, improving accessibility by **90%** and demonstrating strong user-focused design ability.
- Implemented and optimized **ResNet-based deep learning models** using **TensorFlow** and **PyTorch**, achieving **92% accuracy** in chart classification and data extraction.
- Engineered a robust **OCR + NLP pipeline** (Pandas, NumPy, spaCy, Tesseract) to extract titles, axes, legends, and values with **85% precision**.
- Integrated **Text-to-Speech (TTS)** modules, enabling real-time screen-reader-compatible audio feedback and improving interpretation speed by **60%**.
- Presented findings at **ACM W4A '24**, demonstrating strong communication, research skills, and contributions to **accessible AI** and digital inclusion.

Agentic AI Resume Chatbot with RAG – Personal Portfolio Website

[Link for Chatbot](#)

- Built a **production-ready agentic AI chatbot** using **Retrieval-Augmented Generation (RAG)** with a React frontend and FastAPI backend, delivering resume-grounded, hallucination-free responses.
- Implemented a complete RAG pipeline with ChromaDB vector search, SentenceTransformers embeddings, LLM tool/function calling, session-based memory, and GPT-4o-mini inference, enforcing strict grounding and deterministic fallback behavior.

CERTIFICATIONS

AWS Certified Developer – Associate

[Click here to View](#)

Applied Data Science – IBM (Powered by SmartInternz)